Azure Services im eigenen Rechenzentrum – mit und ohne Azure Local, wie funktioniert das?

Alexander Ortha
Cloud Solution Architect Hybrid & Sovereign Cloud Lösungen

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Cloud Solution Architect Hybrid & Sovereign Cloud Lösungen / Ex-Microsoft

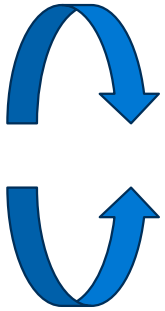
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Azure Local – Transformation from OS into a Solution



Cloud “Arc connected” Operating System





Arc enabled Solution
fully integrated into Azure.
Built as Infrastructure as code from the ground up.

Microsoft Sovereign Cloud

Most comprehensive set of sovereignty solutions across AI, productivity, security and cloud platform

Sovereign Public Cloud

Data stays in Europe, under European law

Operations and access controlled by European personnel

AI model choice and AI processing that stays in Europe

Sovereign Private Cloud

Continuous operations across hybrid or disconnected environments

Integrated cloud and productivity with Microsoft 365 Local

AI capabilities and advanced GPU support with Azure Local

National Partner Clouds

Access in-country expertise and nationally owned organizations

Germany: Delos Cloud designed to meet BSI Cloud requirements

France: Bleu designed to meet SecNumCloud requirements

Consistent infrastructure management and development platform

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Consistent infrastructure management and development platform



Azure Local brings Azure anywhere

Cloud-consistent infrastructure that runs in your datacenter, at the edge, or fully disconnected, managed through Azure Arc.



Cloud-consistent infrastructure: Same Azure APIs, Azure portal, and automation tools whether in the cloud or on-premises



Azure Local runs anywhere: Data center, edge, or fully disconnected environments



Unified management via Azure Arc: Single control plane for provisioning, security, updates, and monitoring across all locations



Broad workload support: VMs, containers (AKS), AI/ML workloads, Azure Virtual Desktop, IoT operations, and data services

Azure Local

AI-ready infrastructure, anywhere

Connected to Azure region

Disconnected from Azure region

Any app

Defender for Cloud

Azure Monitor

Windows & Linux apps

Virtual apps & desktops

Azure data services

Microsoft 365 Local

GitHub Enterprise Local

Foundry Local OSS AI Models

Frontier IP models*

Enabled by Azure Arc

Azure Local

Managed **Kubernetes**/AKS

General-Purpose **VMs**/IaaS

Core Infrastructure Services: **Compute** | **Storage** | **Networking** | **Availability**

Azure-based **Infrastructure Management**

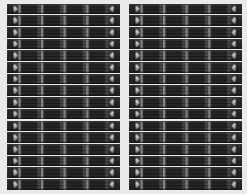
Composable scale point for sovereign AI from single nodes up to multiple racks clusters













* For qualified customers

Azure Local

GA & Preview Features

GA Preview

4-node switchless

Edge RAG

VMConnect

Trusted VM attestation

Domain join pre-deploy

VMware migration

SSH/RDP connect

Terraform templates

GPU acceleration

Data disk expansion

Multi-rack scale

Container Apps on edge

SAN via iSCSI

Local identity · Key Vault

SDN · NSGs

Simplified provisioning

Diagnostics logs

Live VM memory

NVIDIA RTX PRO 6000

Disconnected ops (ALDO)

NVIDIA L-series GPU

Drift detection

Arc Gateway

Update settings

Rack-aware clusters

SAN support

Disaggregated deploy

Live migration GPU-P

Security baseline

Crash dump collection

M365 Local

KAITO on AKS Arc

AI Video Indexer (Arc)

Small Form Factor

Source: Microsoft Learn (What's new / release info) · Azure Arc Blog · Microsoft Ignite 2025 — superseded previews excluded (GA shown)

Built for modern workloads

Azure Local is designed to run the workloads enterprises are building today and tomorrow

Connected and disconnected

Run cloud-integrated or fully disconnected with a local control plane via appliance VM

AI-ready infrastructure

Over 50 GPU-capable platforms with NVIDIA GPUs; supports:

- Edge inferencing
- Video Indexer
- AI Search
- Azure ML workloads

Single Azure control plane

Azure portal, Defender, Monitor, Policy, Update Manager, and Entra ID all extend to Azure Local

Consistent DevOps and IaC

ARM templates, Bicep, Terraform, and GitOps for repeatable, scalable deployments across thousands of locations

Workloads

Infrastruktur

- VMs
- Kubernetes / AKS
- AVD (Windows 11 Multisession)
- M365 Local

AI & DEV

- AI Foundry*
- Github Enterprise*
- Azure IoT Operations
- AI Video Indexer
- AI Custom Vision
- Container Apps*
- Logic Apps Hybrid

* = Preview

Flexible deployment options

Azure Local supports both cloud-connected and fully disconnected deployments with the same platform

Cloud-connected

- Control plane in Azure cloud region
- Consistent connection to Azure for management, security, and updates
- Full suite of Azure services available
- Ideal for data center, edge, and branch scenarios

Run fully disconnected

- Control plane runs locally as appliance VM
- Operates without any connection to Azure public cloud
- Subset of services: Portal, ARM, Registries, Key Vaults, Policy, VMs, Kubernetes
- Ideal for sovereign, defense, and highly regulated environments

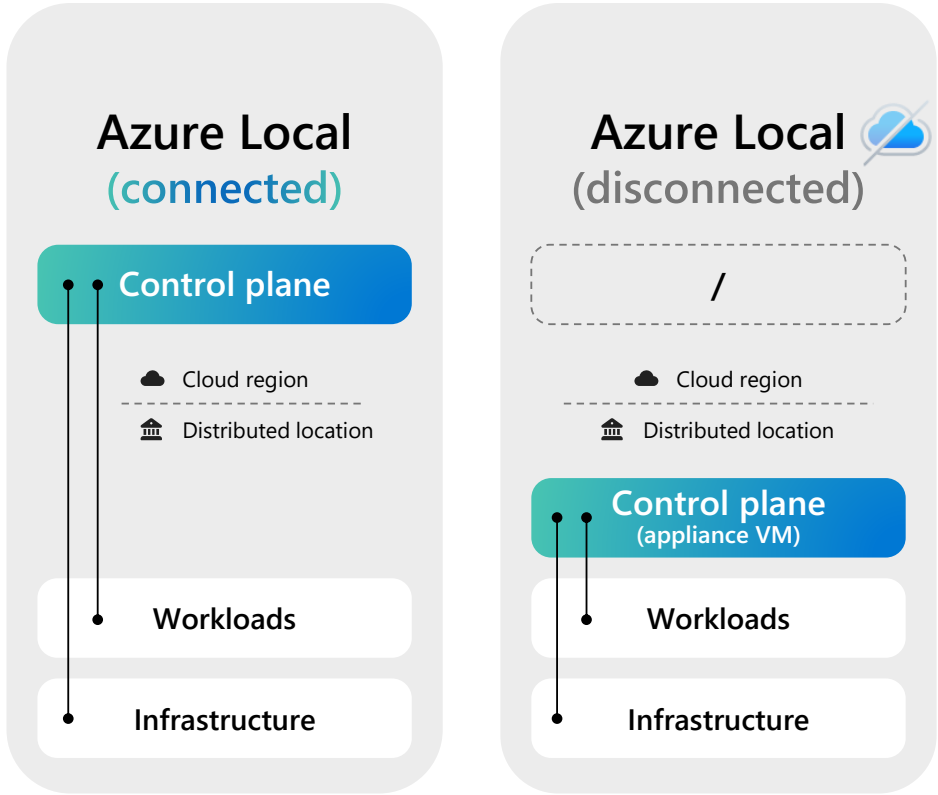
Run fully disconnected

No need to connect to Azure, everything runs on prem.

- Satisfy regulatory requirements by operating permanently disconnected from the cloud
- Host backend Azure resource manager, portal, and services in local appliance VM
- Integrate with existing datacenter systems for Identity, Monitoring, and PKI
- Deploy and manage multiple Azure instances from a single private control plane for workload isolation

Subset of services available:

Portal	ARM	Copilot	Key Vaults	Policy [1]	Defender
Local	Machines	Kubernetes	Registries	AVD	Foundry Local



1 : Policy enforcement. Compliance coming.

Flexible deployment options

Azure Local supports both cloud-connected and fully disconnected deployments with the same platform

Cloud-connected
Hyperconverged

- Control plane in Azure cloud region (for all variants)
- Clusters of 1–16 machines
- Storage Spaces Direct-based virtualized storage (optional SAN)
- Full suite of Azure services available

Cloud-connected
Disaggregated

- Up to 64 machines (1-64)
- Separating compute and storage for independent scaling using SAN storage
- SAN storage from a wide range of partners
- Full suite of Azure services available
- (2605) SAN integration using iSCSI (preview)

Cloud-connected
Multi-rack

- Each instance has one main rack for network aggregation and SAN storage plus three or more compute racks
- The minimum footprint is four racks
- Prescriptive hardware procured from a Microsoft hardware partner
- Full suite of Azure services available

Cloud-connected
Small form factor

- Azure Local on compact, Linux-based edge hardware
- Single-node & small-scale
- Containers (Docker), AKS, K3s, IoT Operations, Foundry Local
- Edge/IoT, retail, manufacturing, device fleets

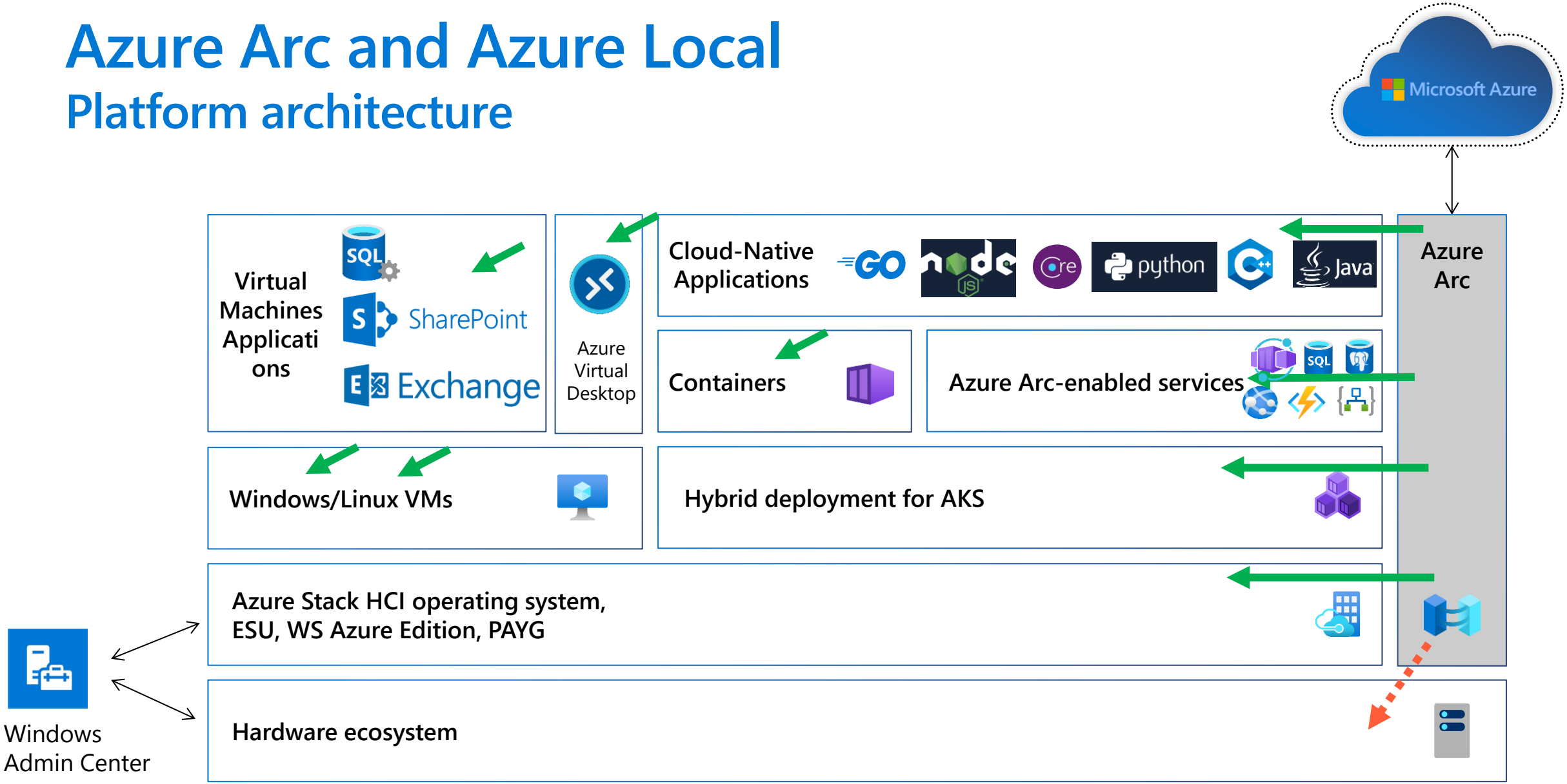
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Control plane / API in Azure cloud region

Azure Arc and Azure Local

Platform architecture



Microsoft Azure

Single control plane with Azure Arc

Infrastructure

Connect and operate hybrid resources as native Azure resources

Azure Arc-enabled infrastructure

Services

Deploy and run Azure services outside of Azure while still operating it from Azure

Azure Arc-enabled services


Servers


K8s


Windows Server



SQL Server


Linux


Microsoft Copilot in Azure


Microsoft Defender for Cloud


Azure Monitor


Azure Update Manager


Azure Policy


Multi-cloud


Datacenter


Edge

Enabled by Azure Arc

Operate with AI-enhanced central management & security

Configuration management & governance

- Azure Update Manager
- Configuration Management
- Azure Policy
- Inventory Management

Resiliency and observability

- Azure Monitor
- Chaos Studio
- Site Recovery

Built-in security and control

- Microsoft Defender for Cloud

Universal AI assistant, portal and tools

- Azure Copilot
- Azure Portal
- Powershell And CLI

Azure Services across your infrastructure

Azure

- Windows Server
- SQL
- Kubernetes
- Linux

Multi-cloud

- aws
- Google Cloud
- ORACLE

On-premises

- PROXMOX
- vmware by Broadcom
- NUTANIX
- Azure Local
- Microsoft Hyper-v

Arc-enabled Infrastructure

Configuration management | Observability | Governance | Security



Microsoft Copilot for Azure



Microsoft Defender for Cloud



Azure Monitor



Microsoft Sentinel



Azure Policy



Azure Update Manager



Configuration Management



Inventory Management

Azure Services across your infrastructure









Azure Local











Azure IoT







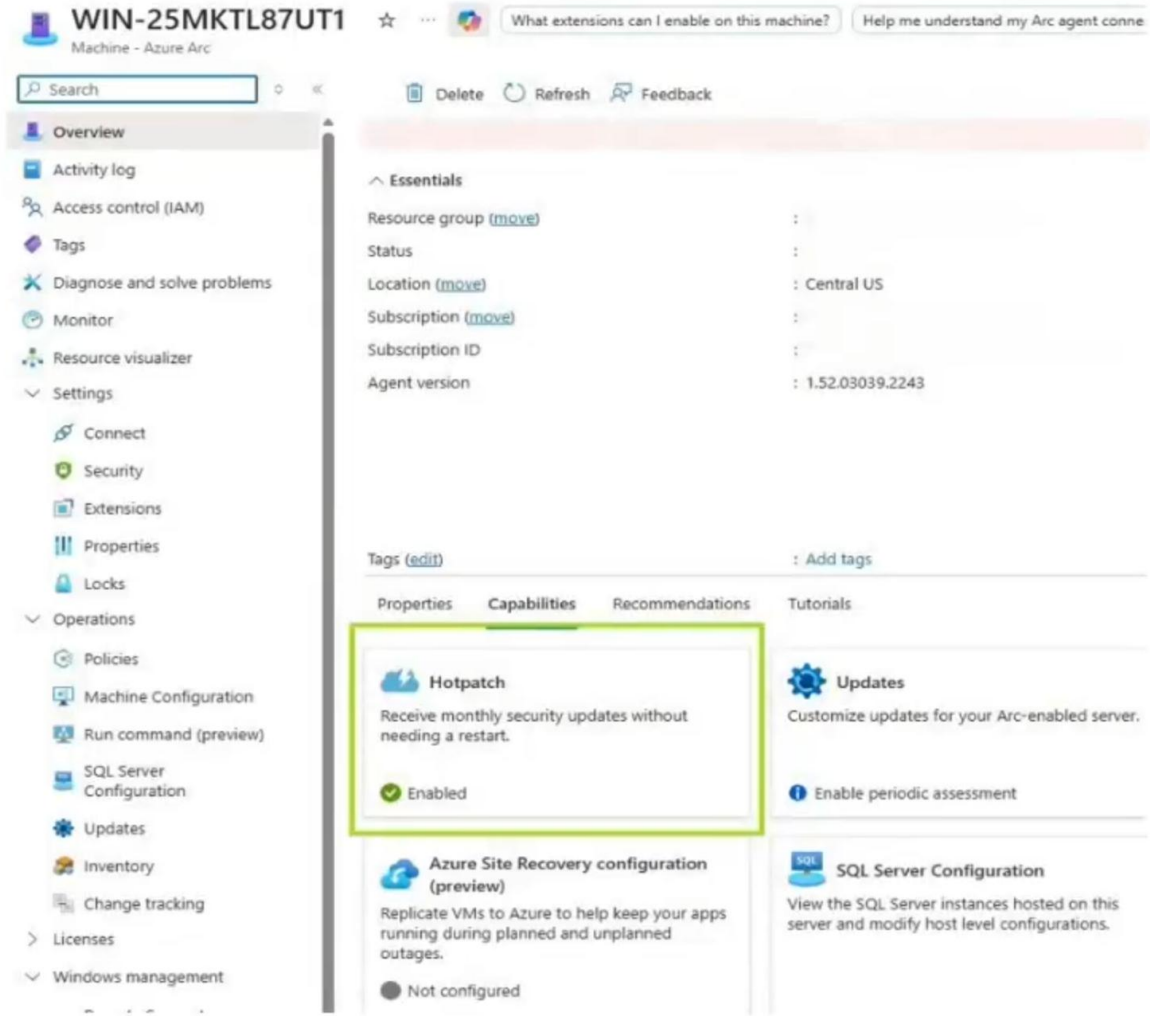
Datacenter, multicloud, and edge

Special Announcement: Hotpatch enabled by Azure Arc

Centralized management of updates across hybrid and multicloud environments

- Minimize downtime
- Accelerate protection
- Unify patch orchestration across machine fleet

...now at **no additional cost** for Windows Server 2025!



ESUs enabled by Azure Arc

Run end-of-support operating systems.

Ensure consistent Windows Server 2012/R2, SQL Server 2012/2014, Windows Server 2016, and SQL Server 2016 performance with high availability and visibility over your entire data and server estate. Keyless delivery.

Security and compliance

Seamlessly extend Azure security and governance to your environment and stay compliant with supported software.

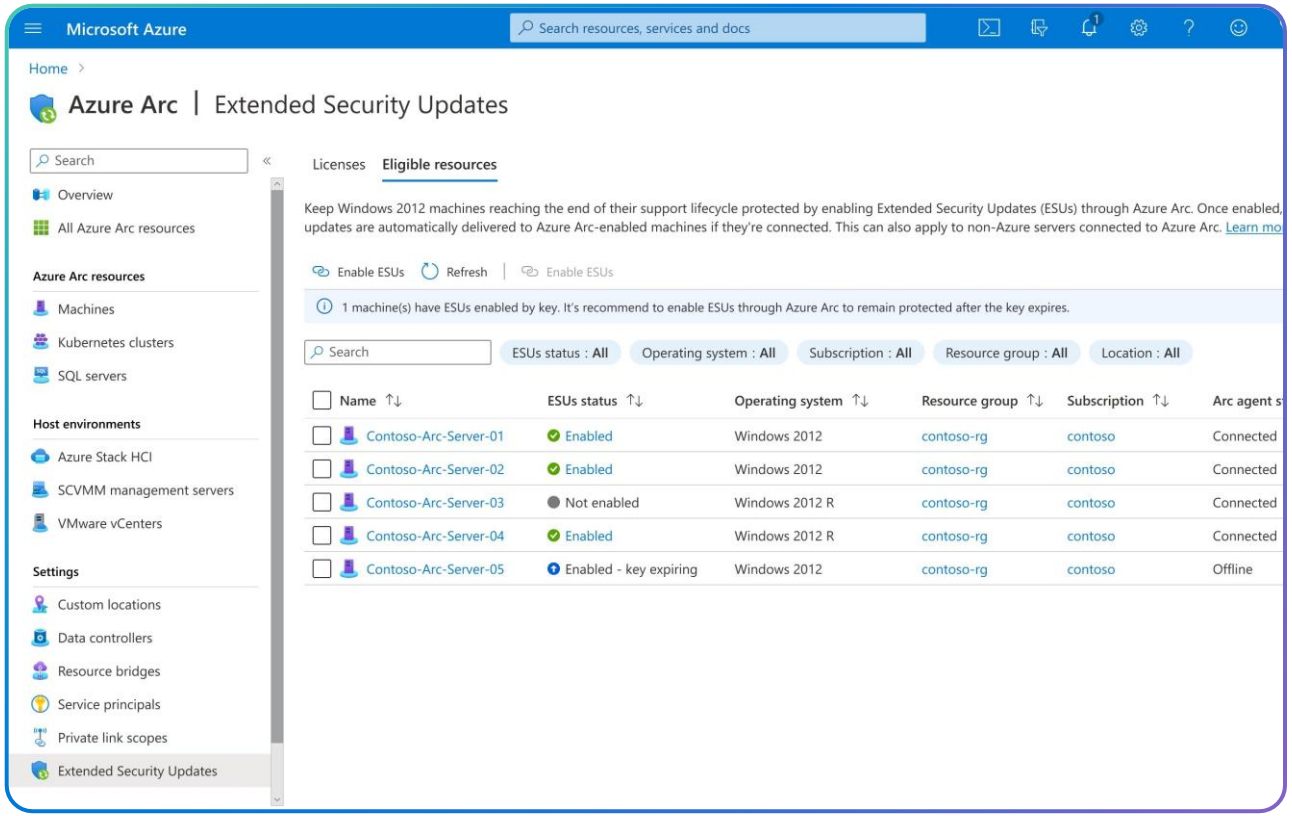
Microsoft Defender for Cloud

Azure Update Manager

Microsoft Sentinel

Azure Policy

Azure Monitor



Enroll and purchase ESUs directly in the Azure Portal

For Windows Server 2012/2012 R2, SQL Server 2012 and SQL Server 2014 (July 2024)
Windows Server 2016 (January 2027), and SQL Server 2016 (July 2026)

Leverage Azure Arc for Server Inventory Management

Change Tracking & Inventory

Unified Resource Management

Once connected to Azure Arc, machines appear as **Azure Resource Manager (ARM) resources.**

Centralized Visibility

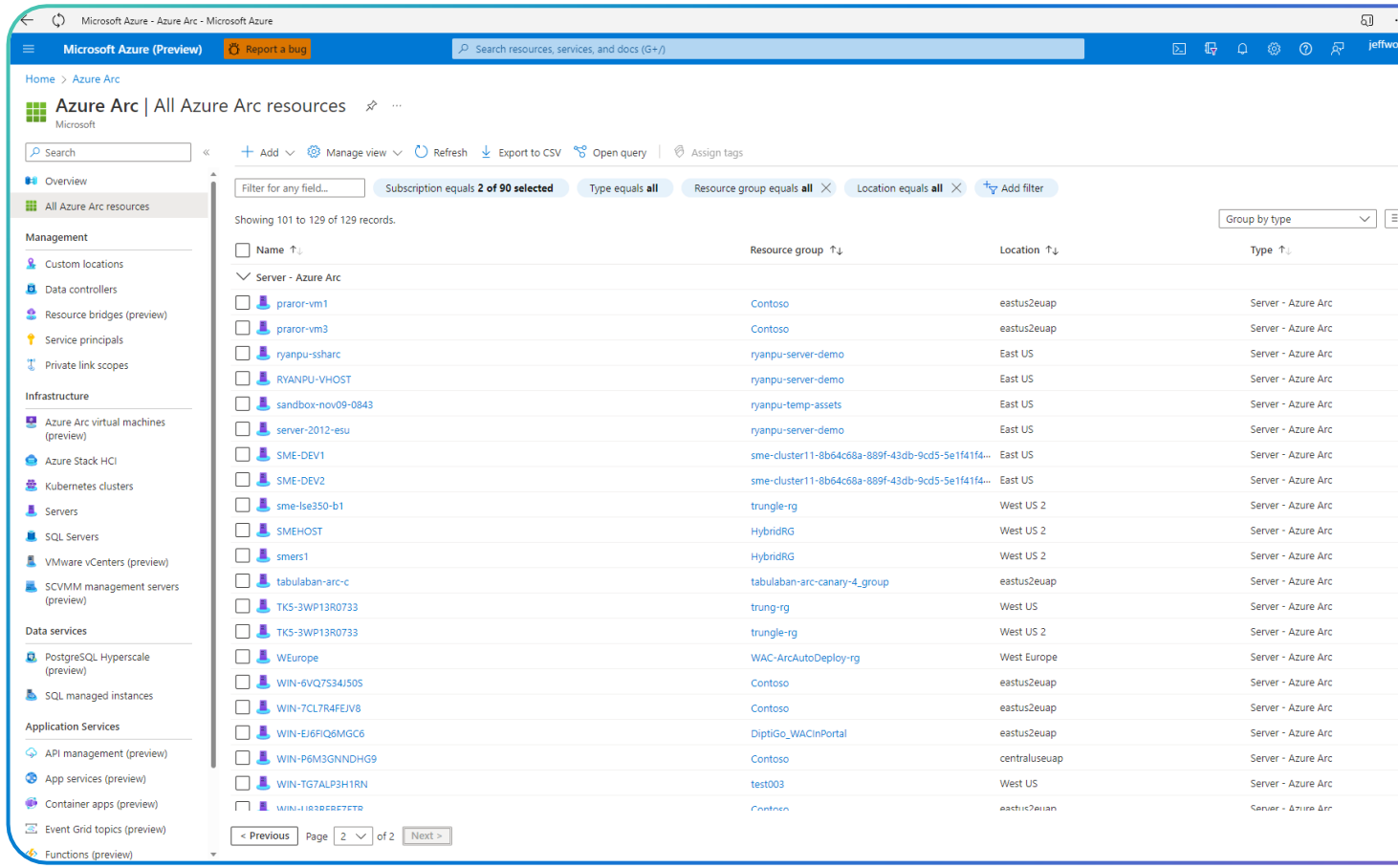
View all connected machines in **the Azure portal**, regardless of location (on-prem, multi-cloud, or edge).

Metadata Access

Access key machine metadata like hardware instance details, cpu utilization, and location directly within Azure.

Resource Organization

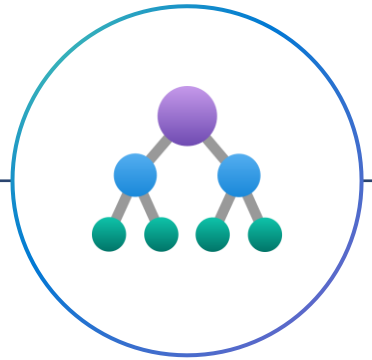
Manage and organize resources using tags.



Governance Services

On by default, Azure governance gives you visibility and control into your Azure Environment.

Management Groups



Streamline management by grouping and controlling Azure subscriptions with ease.

Azure Policy



Ensure compliance and governance in the cloud with automated, enterprise-wide policy enforcement.

Tags



Organize and track your Azure resources effortlessly with customizable, dynamic tags.

Azure Resource Graph



Unlock powerful insights at scale with lightning-fast querying across all your cloud resources

Azure Policy

Active control and governance at scale for your Azure resources

Apply policies at scale

- Apply policies to the root Management Group with control across your entire organization
- Grouping through initiatives
- Exemptions and safe-deployment tools

Enforcement & compliance

- Real-time policy evaluation and enforcement
- Periodic & on-demand compliance evaluation
- Enforcement of configuration and resource actions

Remediation

- Remediate existing resources at scale
- Automatic remediation resources at deployment time
- Trigger Functions, Logic Apps, or web hook upon compliance state changes (via Event Grid)

Machine Configuration

Audit and configure OS, app, or workload level settings as code



Dynamically assign configurations to windows & Linux machines running in Azure and anywhere else via Arc enabled server.



Continuous auditing and enforcement of configuration settings with at scale and granular reporting.



Deploy built-in content provided by Microsoft including security best practices within the Azure Security Benchmark.



Bring your own best practice configurations into Azure through custom artifact deployment and PowerShell DSC.

Management

- Extend Azure Policy to the guest operating system
- Develop configuration as code pipelines leveraging ARM, Bicep, Terraform

Azure Monitor anywhere enabled by Azure Arc

Consistent monitoring across your hybrid and multi-cloud compute

Detect and diagnose issues across **apps and dependencies** with application insights

Correlate issues at **infra level** with insights for VMs, containers, network, etc.

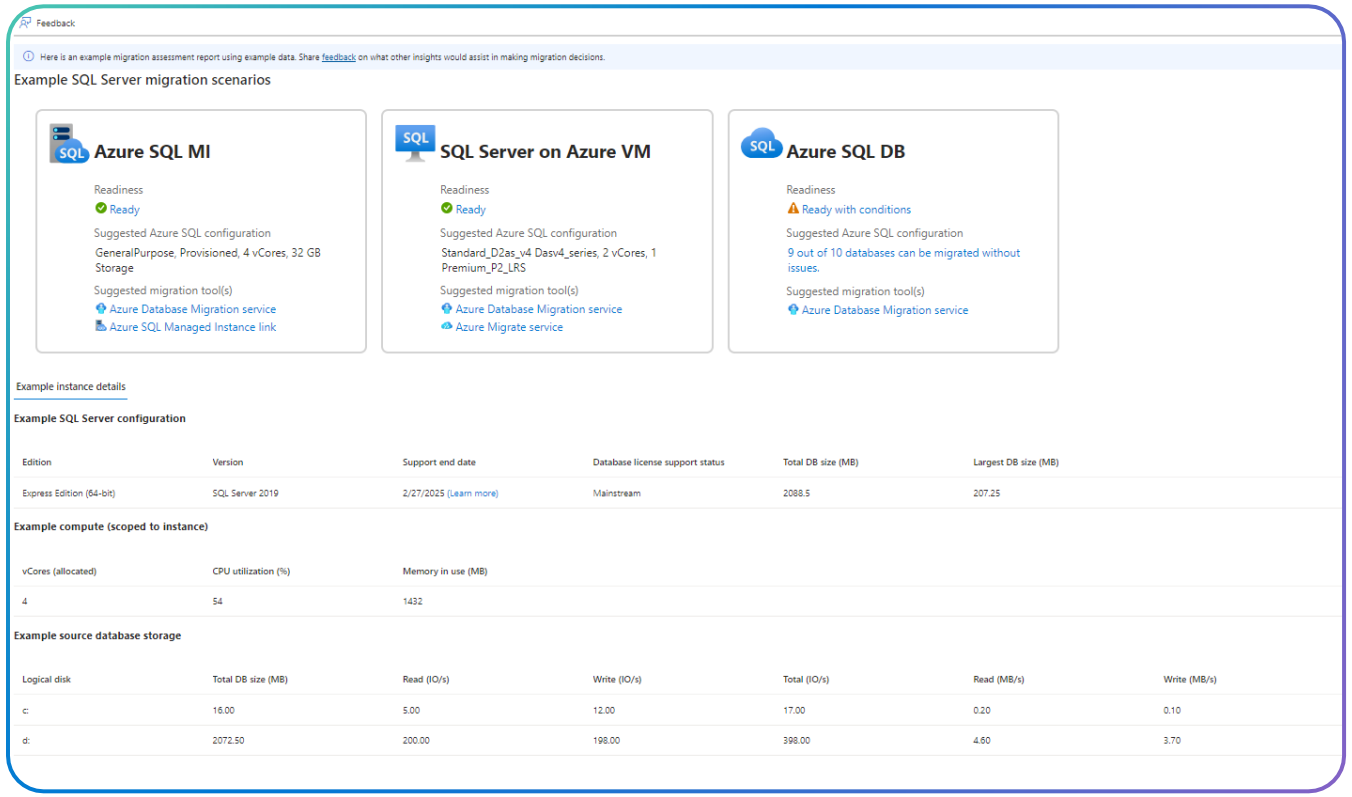
Operationalize at scale with smart **alerts** and automated **actions**

Drill down with **Log Analytics** for troubleshooting & deeper diagnostics

Create **visualizations** with Azure dashboards, workbooks, and Grafana



SQL migration assessment in Azure Arc



Free to all customers running SQL Server anywhere
Available for all SQL Server editions, SQL Server 2012 and above

Enabled by default

Assessment is baked in when customers enable SQL Server with Azure Arc.

Comprehensive and actionable assessment

Evaluate and measure the readiness of SQL Servers for Azure SQL, identify migration risks with mitigation actions.

Best-fit recommendations on SQL IaaS/PaaS

Optimize for performance and cost with guidance on service tier, sizing and migration tooling.

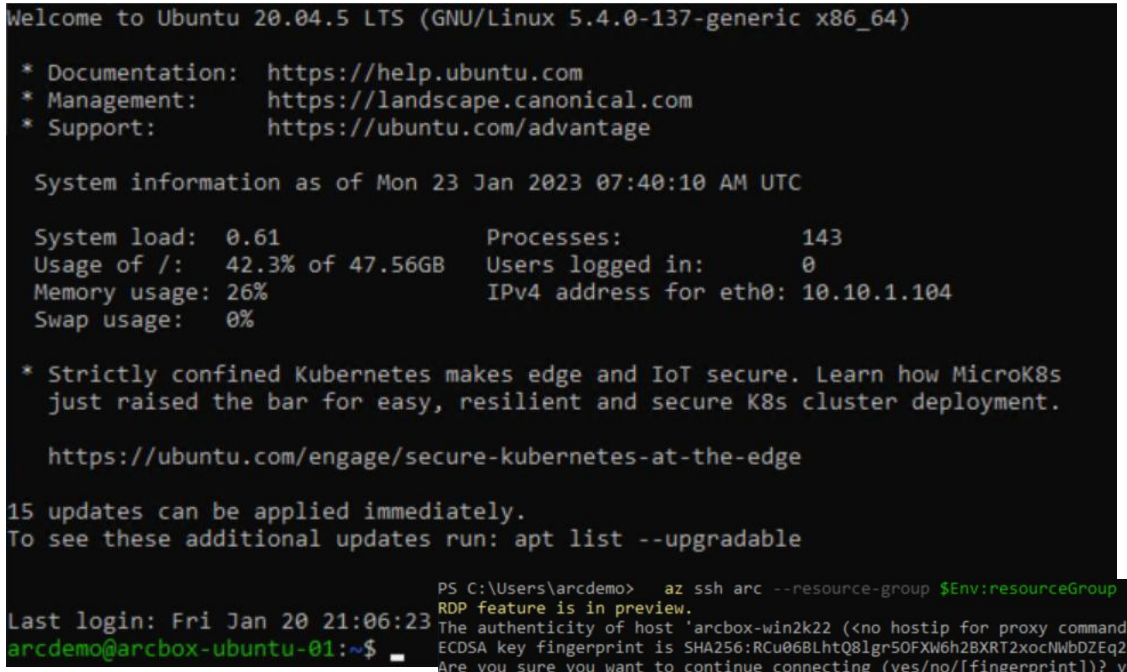
Out of box and continuous reporting

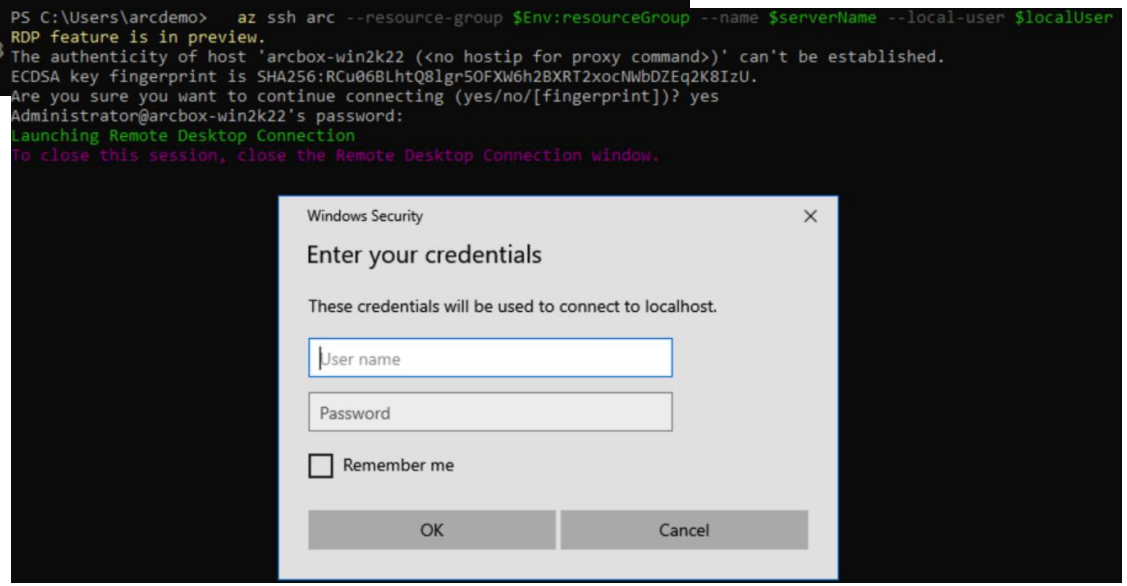
Discovery of SQL Server and its readiness reports are instant, automatic and continuous, with zero additional setup time.

SSH into your Azure Arc-enabled servers

How to SSH into your Arc-enabled servers:

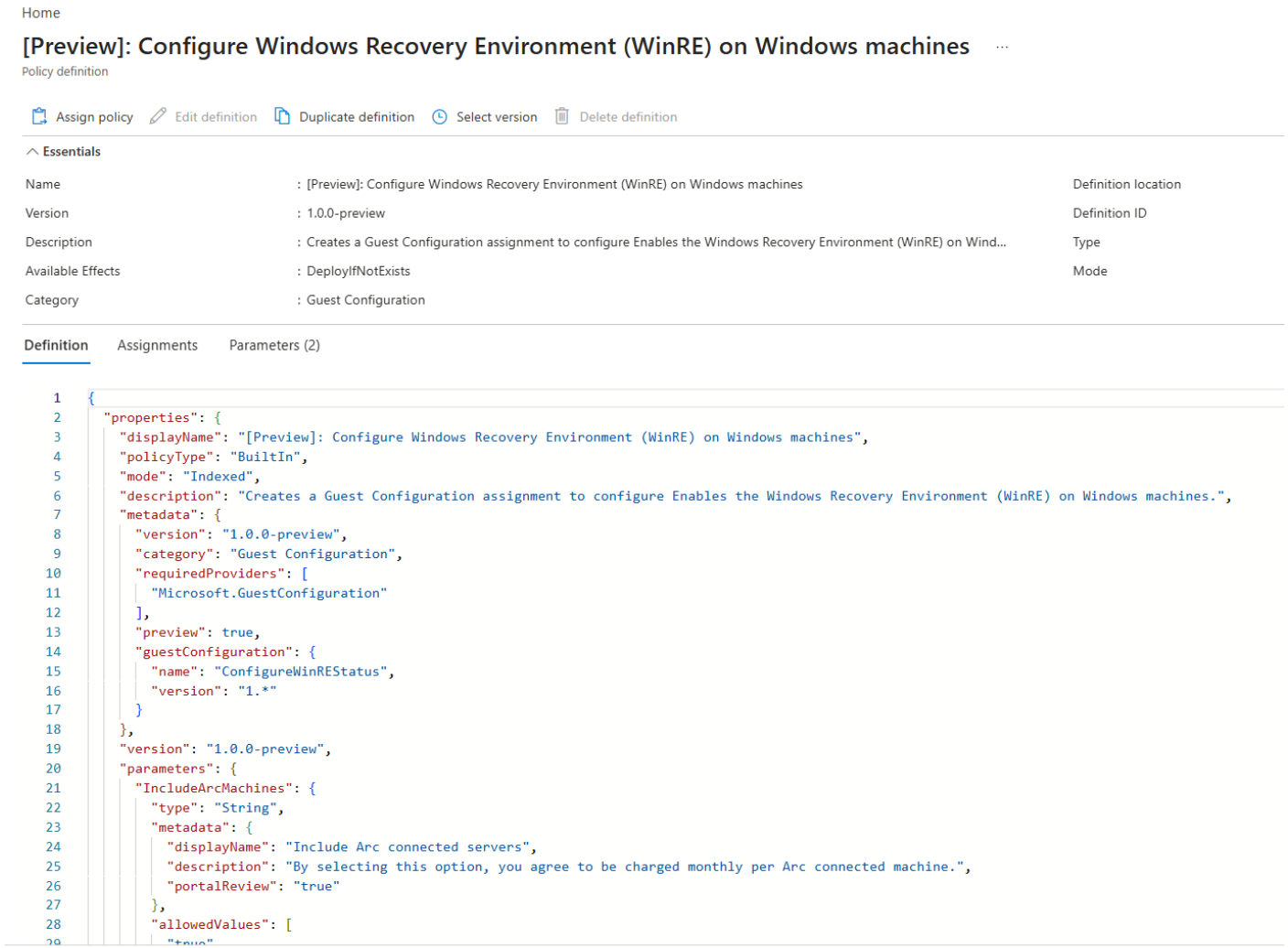
- Install Azure CLI SSH extension or/and Azure PowerShell SSH module on the client machine
- Enable SSH service on Arc-enabled servers
- Connect to Arc-enabled servers using SSH or using Remote Desktop tunneled via SSH
- Connect using Azure AD/Entra ID based SSH login





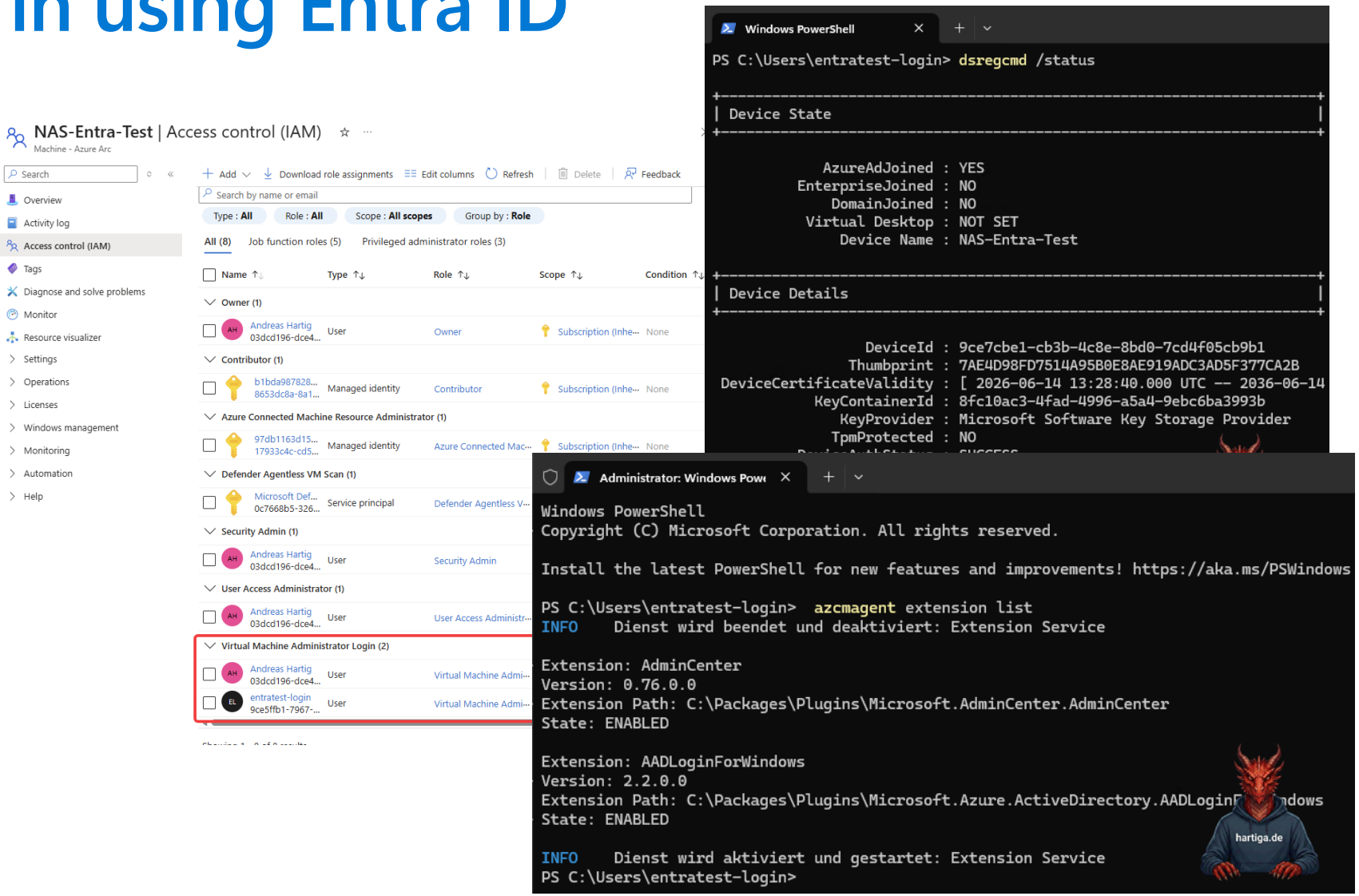
Public Preview: Audit and Enable Windows Recovery Environment (WinRE) for Azure Arc-enabled Servers

[Public Preview: Audit and Enable Windows Recovery Environment (WinRE) for Azure Arc-enabled Servers | Microsoft Community Hub]



Azure Arc – Sign in using Entra ID

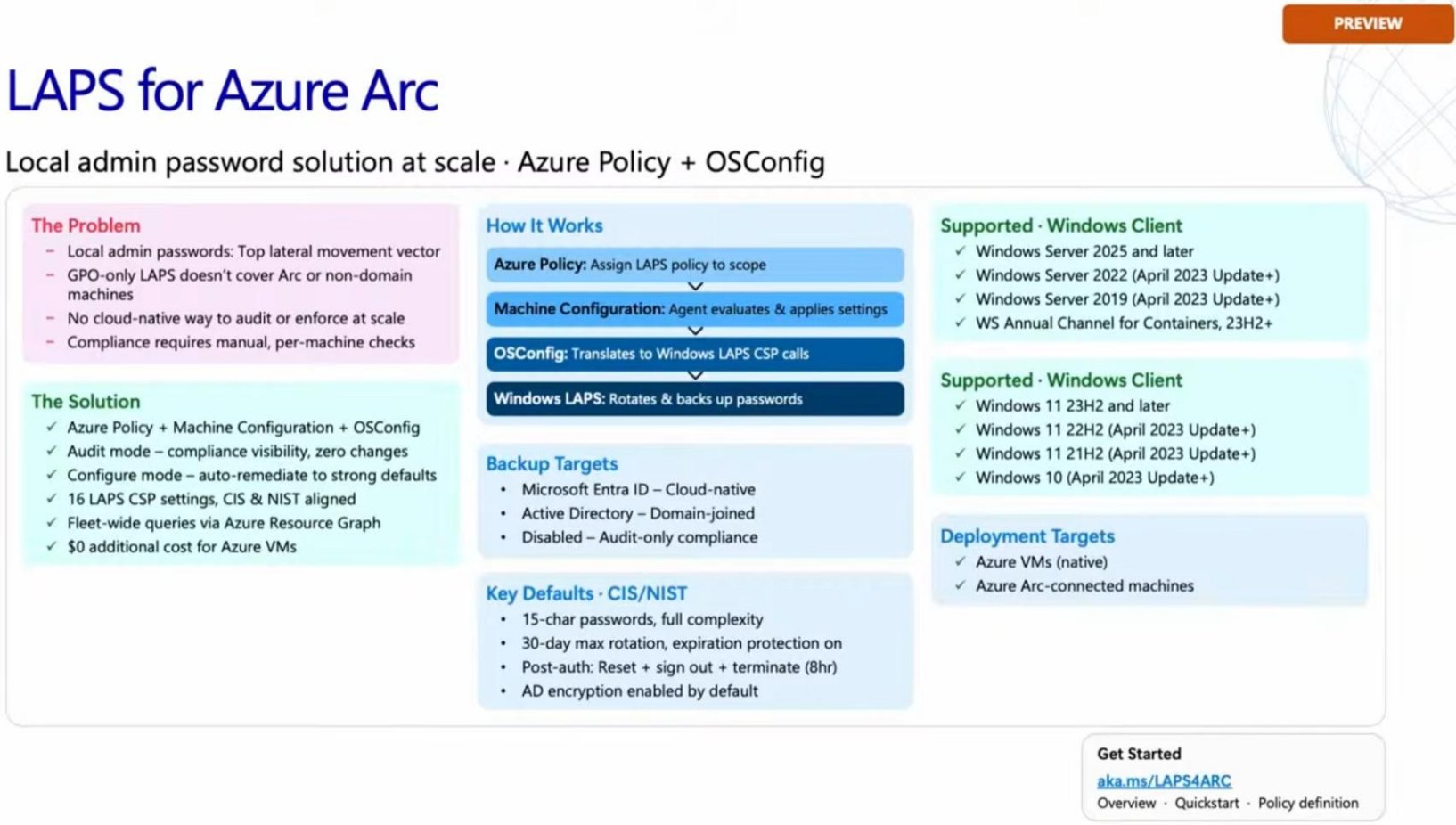
- Enforce MFA for Windows Server
- No requirement for Active Directory
- Use existing Entra ID credentials
- Azure Audit Capabilities vs. „Windows Event Viewer“
- No additional license costs
- Single Control Plane for servers incl. Azure Policies to replace GPOs



[Sign in to an Azure Arc-enabled server using Microsoft Entra ID - Microsoft Entra ID | Microsoft Learn]

Azure Arc – LAPS for Azure Arc (Preview)

- LAPS for Azure Arc enabled machines (Prev
- Cloud Native Passwort Backup
- Neue Azure LAPS Policy
- Azure Policies statt GPOs
- Passwort Rotation
- Windows Server 2019 bis 2025
- Single Control Plane for LAPS



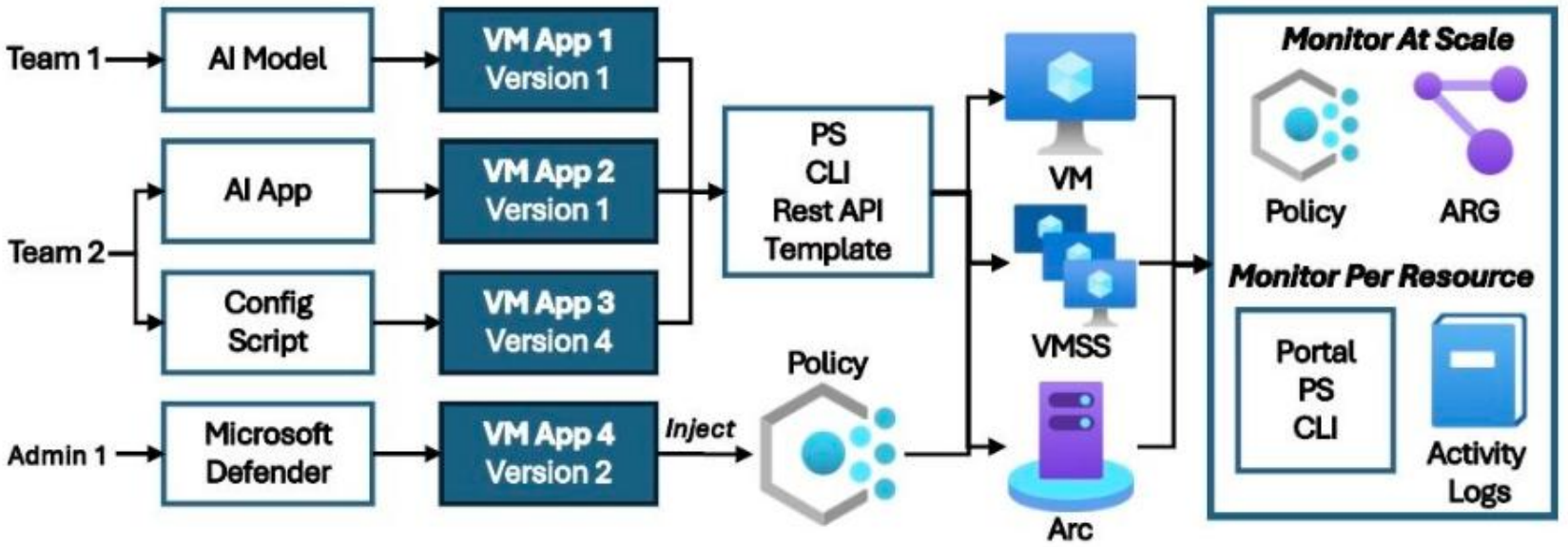
https://aka.ms/LAPS4ARC - Windows Server Summit 2026 "Security baselines, benchmarks, posture, and scale"

VM Applications

Learn More: **Aka.ms/VMApplications**

Managed & one stop solution for VM software lifecycle management

- Faster, modular, scalable & resilient deployments vs scripts.
- Private app gallery with role-based access controls
- Inventory monitoring & governance
- Replica per region for faster creates & resiliency



[Overview of VM Applications in the Azure Compute Gallery - Azure Virtual Machines | Microsoft Learn]

[Software Management on Azure Arc - Private Preview Signup]

Danke für die Aufmerksamkeit – Fragen?

Alexander Ortha

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